

Organize Spice Rack SOP (12x Jars)

One episode is one full rack of twelve jars. It runs from twelve jars standing on a three tier rack in mixed order on dusty shelf strips, to the same twelve jars back on the rack: strips wiped, jars tiered by height, ordered alphabetically by label within each tier, and every label facing forward.

The right gripper does every step. The left arm stays at home for the whole episode and never enters the workspace. Nothing is ever handed between grippers and only one object is ever held. The right arm is the working arm because the rig is set up right handed, so the dominant hand drives it in leader follower. A left handed collection requires a separately approved mirrored variant that swaps the arm assignments and workspace. Do not mirror this SOP or swap arms during an episode.

The order never changes: **unload all twelve jars, wipe the three shelf strips, sort the jars in the staging area, load the tiers back to front, then face every label forward.** No jar ever goes back onto a strip that has not been wiped.

Never move more than one jar per pick. Never drag or slide a jar during transport: every jar is lifted clear of the strip and carried. A small in-place rotation is permitted only when reading a label in the staging area or facing a label on the rack. Never lay a jar on its side or tip it past a slight lean. Never grip a jar by its cap. Never lift a jar to turn it in the air. Never push, lean on, or steady the rack with the gripper: it stands where it was set for the whole episode.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the rack and its frame includes all three tiers, the staging area, and the cloth park.
3. Every label is legible on camera when a jar is faced, and the dust or spill on each shelf strip is visible before the wipe.
4. The right arm is at home with its gripper open.
5. The left arm is at home, clear of the arc, and stays there for the whole episode.
6. The tabletop is clear of anything but the rack, the twelve jars, and the cloth.
7. The right arm reaches every slot on all three tiers, the whole staging area, and the cloth park without extending to a joint limit or folding in on itself. Move the rack forward or narrow the staging area until this is true.
8. The arm's path along the arc is unobstructed, and the back tier can be approached from above without the arm passing through the middle or front tier.

Materials checklist

1. The rack stands at the inboard end of the arc, set back and square to the front table edge, with four slots on each of its three tiers.
2. The rack stands steady on its own. It does not rock, slide, or tip when a strip is wiped or when a jar is set into a slot. Weight it or replace it if it moves.
3. All twelve jars stand on the rack, four per tier, caps on tight.
4. The jars are in mixed order: heights are not tiered and labels are not facing forward.
5. The jar set is four tall, four medium, and four short, with every label legible and the three heights clearly different by eye.
6. Every shelf strip carries visible dust or spill for the wipe to remove.
7. The staging area is empty and wide enough to hold three separated groups of four jars.
8. The folded cloth lies flat and dry on the cloth park.

Workspace layout

One arc, three zones, all on the right arm's side of the table and all inside its comfortable reach. Reading from the inboard end (toward the middle of the table) to the outboard end:

1. **Rack position** (inboard end), set back and square to the front table edge. The rack is both the input and the output, and it stays where it was set for the whole episode.
2. **Staging area**, outboard of the rack, holding the twelve jars between the unload and the load.
3. **Cloth park** (outboard end), where the folded cloth rests when it is not in use.

Jars travel outboard from the rack to the staging area and inboard again on the load. No carry passes over a loaded tier or over a jar already standing in the staging area.

Vocabulary

- **Arm assignment:** the right gripper performs every action in this SOP. The left arm never leaves home, never enters the workspace, and never steadies a jar, the cloth, or the rack.
- **Jar:** one capped spice jar to be sorted and placed. **Tall**, **medium**, and **short** jars are the four tallest, the four mid height, and the four shortest in the set.
- **Label:** the printed face carrying the spice name. This is the face that must end up forward.
- **Shelf strip:** the flat run of one tier that the jars stand on. There are three: the **back tier** is the highest and rearmost and takes the tall jars, the **middle tier** takes the medium jars, the **front tier** is the lowest and frontmost and takes the short jars.
- **Slot:** one of the four standing positions along a shelf strip, counted left to right from the front of the table.
- **Label order:** alphabetical order of the label text, read left to right. Compare the first letter, then the second, then the third. Two identical labels may go in either order.

- **Faced:** the jar stands upright with its label square to the front of the table and readable in the environment camera. A label turned to a side or to the back is not faced.
- **Cloth:** the folded cloth used on the shelf strips. It is held so a broad flat side of it meets the strip, never by a corner and never balled up.
- **Pass:** one continuous draw of the cloth along the full length of one shelf strip, left to right, in contact with the surface from end to end.
- **Release point:** the gripper opens only in a slot on a wiped strip, in the staging area, or on the cloth park. Never over a loaded tier, over another jar, or over open tabletop.
- **Stable placement:** the jar stands still after release, fully on the surface, clear of the strip edge, and does not rock, lean on a neighbour, or slide.
- **Spill:** dust or debris that lands on the tabletop during the wipe. It is left where it falls and cleared in the reset, never on camera.

Steps

Every step below is performed by the right gripper.

Step 1: Unload the twelve jars

Goal: all three shelf strips are empty and the twelve jars stand separated in the staging area.

Work one jar per pick, front tier first, then the middle tier, then the back tier, so no reach passes over a loaded tier.

- Approach the leftmost jar on the front tier from above and close on the jar body below the cap, with the gripper square to the jar so it hangs upright rather than tilted.
- Lift it straight up clear of the shelf strip, carry it outboard to the staging area, and set it down upright.
- Release once it stands on its own, clear of the jars already there and clear of the table edge.
- Work left to right across the front tier, then the middle tier, then the back tier, one jar at a time, until the rack is empty.

Never close on the cap, never lift two jars in one pick, and never slide a jar along the strip to get at it.

Check: all three strips are clear and the twelve jars stand separated in the staging area, touching neither each other nor the table edge. If a jar remains on a tier, take it now. If a staged jar is leaning or touching another, settle it before continuing.

This secure-grip rule applies to every jar pick in Steps 1, 3, and 4: if a grip is not secure before lifting, open and re-approach. Make no more than two corrective takes on one jar; if it still cannot be held securely, end the episode and inspect it during reset.

Step 2: Wipe the three shelf strips

Goal: all three strips are wiped clear of dust and spill, the rack has not moved, and the cloth is back on its park.

2.1 Take the cloth

- Approach the cloth on its park from above and close on it so it is held flat, with enough of it below the gripper to meet the strip.

Check: the cloth presents a flat face and does not hang by a corner or slip. If it does, set it down on the park and take it again. Make no more than two corrective takes; if it is still not secure, end the episode and correct the cloth or setup during reset.

2.2 Wipe back to front

- Bring the cloth down onto the back tier strip at its left end and draw it along the full length of the strip, left to right, keeping contact for the whole pass.
- Lift at the end of the pass, then run the same pass on the middle tier, then on the front tier.
- Press only hard enough to clear the surface. Draw the cloth along the strip, never into the rack, and never lift or lean on a tier.

Wipe back to front so debris is never pushed onto a strip that has already been wiped. Do not lift the cloth part way through a pass.

Check: after each pass the full length of that strip is clear, including both ends and the back corner. If a patch remains, repeat that pass before moving on. If the rack has shifted, end the episode rather than moving it back.

2.3 Park the cloth

- Carry the cloth to the cloth park, lay it down flat, release, and lift clear.

Check: the cloth lies fully on the park, not draped over the rack, a jar, or the staging area. The gripper is empty before any jar is handled again.

Step 3: Sort the jars in the staging area

Goal: the twelve jars stand as three groups of four, tall, medium and short, each group already in label order.

Sorting on the table is cheaper than correcting a loaded tier, so the order is settled here.

3.1 Group by height

- Move jars one at a time until the four tall jars stand together, the four medium jars stand together, and the four short jars stand together.
- Keep a clear gap between the three groups so the grouping is readable on camera, and stand each jar upright before reaching for the next.

Check: four jars in each group. If a group holds three or five, the height call is wrong. Compare the jars side by side and move the odd one before continuing.

3.2 Order each group by label

- Read the labels in the tall group and move jars one at a time until they stand in label order, left to right.
- Do the same for the medium group, then the short group.
- If a label is not visible, close lightly on the standing jar and rotate it in place only far enough to read it. Do not lift it or translate it across the staging area while turning it. Final facing is still performed on the rack in Step 5.

Check: each group reads alphabetically left to right before the load begins.

Step 4: Load the tiers back to front

Goal: all twelve jars stand on the rack, tiered by height, in label order left to right on each tier.

Load the back tier first, then the middle tier, then the front tier, so no reach passes over a loaded tier.

- Take the leftmost jar of the tall group and place it upright in the leftmost slot of the back tier.
- Work left to right, filling the four slots of the back tier in the same order the tall group stands in.
- Set each jar down fully on the strip, clear of the strip edge and not leaning on its neighbour, and release only once it stands on its own.
- Fill the middle tier with the medium group the same way, then the front tier with the short group.

Approach every slot from above and clear of the tier behind it. Do not brush a loaded jar on the way in or out, and do not slide a jar into position once it is down.

Use no more than two small placement corrections per jar.

Check: tall on the back tier, medium on the middle tier, short on the front tier, four per tier, each tier reading alphabetically left to right. If a jar is on the wrong tier or out of order, correct it now, before facing.

Step 5: Face every label forward

Goal: all twelve jars are faced, with every label square to the front of the table and readable on camera.

- Close lightly on the first jar of the back tier without lifting it, and turn it where it stands until its label is square to the front.
- Release and move to the next jar. Work the back tier left to right, then the middle tier, then the front tier.

Use no more than two small facing corrections per jar. The written wrist reset below does not count as an additional correction.

If a label needs more turn than the wrist has left in one motion, open the gripper, return the wrist, close on the jar again, and finish the turn. Never lift the jar and spin it in the air, and never slide it along the strip while turning it.

Check: every label is square to the front, no jar has moved off its slot or out of label order, and every cap is still on.

Step 6: End the episode

- Confirm the wipe: no dust or spill line is visible on any shelf strip.
- Confirm the tiering: four tall on the back tier, four medium on the middle tier, four short on the front tier.
- Confirm the label order: each tier reads alphabetically left to right.
- Confirm the facing: all twelve labels are square to the front.
- Confirm the staging area is empty and the cloth lies flat on its park.
- Correct a failed check only with the correction written in its step. If the correction would require reversing the required sequence or no correction is written, end the episode and reset instead.
- Return the right arm home with the gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the twelve jars off the rack and stand them back on it in mixed order: heights not tiered and labels not facing forward.
2. Tighten any cap that has loosened, and replace a jar whose label is worn or unreadable.
3. Put dirt back on each shelf strip so the next episode has something to wipe.
4. Fold a fresh dry cloth flat and lay it on the cloth park. Replace a cloth that is damp, shedding, or loaded with dust.
5. Stand the rack back at the inboard end of the arc, square to the front table edge, and confirm it does not rock or slide.
6. Clear any spill off the tabletop and leave the staging area empty.
7. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Left arm used

- **Visible cue:** the left arm leaves home at any point, enters the workspace, touches a jar, the cloth, or the rack, steadies the rack, or assists with a step.
- **SOP rule broken:** Steps 1 to 5, the right gripper performs every action and the left arm never leaves home.
- **Coaching note:** one arm does the whole task. The left arm stays parked.

Violation: Required sequence not followed

- **Visible cue:** the wipe starts before the rack is empty, a jar goes back on before the wipe is finished, the load starts before the sort is done, the facing starts before the load is done, or a step is skipped.
- **SOP rule broken:** Steps 1 to 5 (unload, wipe, sort, load, face, in that order).
- **Coaching note:** empty it, wipe it, sort it, load it, face it. No step out of turn.

Violation: Jar placed on an unwiped strip

- **Visible cue:** a jar goes onto a tier that has not been wiped yet, or the load begins before Step 2.2 is finished.
- **SOP rule broken:** Step 2 (all three strips are wiped before any jar goes back on).
- **Coaching note:** the wipe always comes first. A jar set on a dusty strip means the whole strip has to be cleared and loaded again.

Violation: Rack not fully unloaded before the wipe

- **Visible cue:** the wipe starts with one or more jars still standing on a tier, or the cloth wipes around a jar.
- **SOP rule broken:** Step 1 (all three strips are clear before the wipe begins).
- **Coaching note:** unload all twelve first. Wiping around a jar leaves a ring and risks knocking it over.

Violation: Shelf strip only partly wiped

- **Visible cue:** a pass covers only part of a strip, skips an end or the back corner, the cloth lifts part way through, or a tier is skipped and visible dust or spill remains.
- **SOP rule broken:** Step 2.2, one continuous pass along the full length of each strip, checked before moving on.
- **Coaching note:** one continuous pass per strip, end to end, cloth in contact the whole way.

Violation: Wiped in the wrong direction

- **Visible cue:** the strips are wiped front to back, or in mixed order, so debris is pushed onto a strip that has already been wiped.
- **SOP rule broken:** Step 2.2 (wipe the back tier, then the middle tier, then the front tier).
- **Coaching note:** always wipe back to front so nothing lands on a finished strip.

Violation: Wiped without the cloth, or with the cloth bunched

- **Visible cue:** a strip is wiped with the bare gripper or with a jar, or the cloth is held by a corner or balled up so only a narrow line of it touches the strip.
- **SOP rule broken:** Steps 2.1 and 2.2, the strip is wiped with the cloth held flat and never bunched.
- **Coaching note:** a broad flat face on the strip. A bare gripper smears the spill instead of lifting it.

Violation: Cloth not parked before jars are handled again

- **Visible cue:** Step 3 or Step 4 begins with the cloth still held, or the cloth is left on the table, on the rack, or on a jar instead of flat on its park.
- **SOP rule broken:** Step 2.3 (lay the cloth flat on the park and lift clear before any jar is handled).
- **Coaching note:** park the cloth before you touch another jar. A cloth in the gripper or draped on the rack contaminates the placement.

Violation: Jar placed on the wrong tier

- **Visible cue:** a tall jar ends on the middle or front tier, a medium jar on the back or front tier, or a short jar on the back or middle tier; or a tier ends with three or five jars.
- **SOP rule broken:** Step 4 (tall on the back tier, medium on the middle tier, short on the front tier, four per tier).
- **Coaching note:** tier by height, tallest at the back. Compare the jars side by side in the staging area if a height call is unclear.

Violation: Tiers loaded out of order

- **Visible cue:** the front or middle tier is loaded before the back tier, so an arm reaches over jars already placed.

- **SOP rule broken:** Step 4 (load the back tier, then the middle tier, then the front tier).
- **Coaching note:** fill the rack back to front. Reaching over a loaded tier is how jars get knocked over.

Violation: Label order wrong within a tier

- **Visible cue:** a finished tier does not read alphabetically left to right, or two jars are swapped relative to label order.
- **SOP rule broken:** Steps 3.2 and 4, each tier carries its labels in the fixed order.
- **Coaching note:** settle the order in the staging area, then keep the group order as you place. Read each tier left to right before facing.

Violation: Label not faced forward

- **Visible cue:** at the end of Step 5 one or more labels are turned to a side, to the back, or visibly off square to the front.
- **SOP rule broken:** Step 5, every label square to the front of the table.
- **Coaching note:** face every jar, including the back tier. A label the camera cannot read is not faced.

Violation: Jar lifted or slid to face it

- **Visible cue:** a jar is picked up and turned in the air, or slid along the strip while being turned, rather than turned where it stands.
- **SOP rule broken:** Step 5 (turn the jar where it stands, without lifting or sliding it).
- **Coaching note:** close lightly and turn on the spot. If the wrist runs out of turn, open, return the wrist, and take it again.

Violation: Jar dragged or slid instead of lifted clear

- **Visible cue:** a jar is pushed or slid along a shelf strip or the table instead of being lifted clear and carried.
- **SOP rule broken:** Steps 1, 3 and 4 (lift each jar clear of the surface and carry it).
- **Coaching note:** lift straight up, carry, set down. Dragging scratches the strip and tips the jar.

Violation: More than one jar handled at once

- **Visible cue:** the gripper lifts or moves two or more jars in a single pick.
- **SOP rule broken:** Steps 1, 3 and 4 (one jar per pick).
- **Coaching note:** exactly one jar per pick, every time.

Violation: Jar gripped by the cap

- **Visible cue:** the gripper closes on the cap instead of the jar body, or a cap is unscrewed, knocked off, or left visibly loose.
- **SOP rule broken:** Step 1 (close on the jar body below the cap).
- **Coaching note:** grip the body, never the cap. A jar carried by its cap can come apart mid transport.

Violation: Jar tipped or laid on its side

- **Visible cue:** a jar is carried tilted well off vertical, laid down on the table or a strip, or set down leaning on a neighbour.
- **SOP rule broken:** Steps 1 to 5, each jar stays upright and stands on its own before release.
- **Coaching note:** grip square to the body so it hangs upright, and confirm it stands alone before releasing.

Violation: Jar left overhanging the strip edge

- **Visible cue:** a placed jar is not fully on the shelf strip: its base overhangs the front or the end of the strip.
- **SOP rule broken:** Step 4 (set each jar down fully on the strip, clear of the edge).
- **Coaching note:** aim the base into the slot, not at the edge. An overhanging jar falls on the next pass.

Violation: Jars left touching in the staging area

- **Visible cue:** staged jars stand against each other or in a cluster, so the height groups cannot be read on camera.
- **SOP rule broken:** Steps 1 and 3.1 (stand the jars separated, with the three groups clearly apart).
- **Coaching note:** leave a gap between jars and between groups. The sort has to be visible in the frame.

Violation: Rack pushed, leaned on, or displaced

- **Visible cue:** the rack is visibly pushed, rotated, rocked, or shifted from its setup position, or the gripper closes on the rack to steady it, most often under wipe pressure or an off target placement.
- **SOP rule broken:** Steps 2 and 4, never push, lean on, or steady the rack, draw the cloth along the strip and approach every slot from above.
- **Coaching note:** light pressure along the strip, never into the rack. The rack holds itself.

Violation: Gripper released over the wrong place

- **Visible cue:** the gripper opens over a loaded tier, over another jar, or over open tabletop, or a jar is dropped into a slot from above rather than set down.

- **SOP rule broken:** Steps 1 to 5, the gripper opens only in a slot on a wiped strip, in the staging area, or on the cloth park.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Grip fumbled and taken again

- **Visible cue:** the gripper closes on a jar or the cloth, finds the grip off, opens, and closes again more than twice before lifting.
- **SOP rule broken:** Steps 1 to 4 (approach from above and grasp securely on the first close).
- **Coaching note:** the approach angle is off. Line the gripper up before closing so the first grip catches the jar body squarely.

Violation: Repeated fiddling with a placement or a facing

- **Visible cue:** more than two small corrections are made to settle a jar in its slot or to square a label. The written turn recovery in Step 5 does not count.
- **SOP rule broken:** Steps 4 and 5, settle in one or two small corrections.
- **Coaching note:** grip or approach is off and is forcing correction. Tighten the approach so each jar lands close to target and close to square.

Violation: Jar dropped, collided, or knocked over

- **Visible cue:** a jar falls or is struck, is tipped over on the rack, in the staging area, or on the table, whether by the gripper on approach or retract, by the cloth, or by a neighbouring jar going down.
- **SOP rule broken:** Steps 1 to 5, approach and retract clear, keep jars separated, and release only after a stable placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Spill collected mid episode

- **Visible cue:** an arm stops to pick up, sweep, or push dust and debris that fell on the tabletop back onto a strip or into a jar.
- **SOP rule broken:** Steps 1 to 5, spills are left where they fall and cleared in the reset.
- **Coaching note:** keep going. Spills are cleared in the reset, not on camera.

Violation: Required check not followed

- **Visible cue:** a failed check is visible and the episode continues without the written correction, or a recovery not written in the SOP is used.
- **SOP rule broken:** Steps 1 to 5, perform each check and use only its stated correction.
- **Coaching note:** stop at each check and use only its written recovery.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a failed check uncorrected, a jar left in the staging area, the cloth off its park, the arm close to home but not at it, the gripper not fully open.
- **SOP rule broken:** Step 6 (confirm the wipe, the tiering, the label order, the facing, and the staging area and cloth park; return the arm home; then stop recording).
- **Coaching note:** confirm, hold, home, stop.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective rack:** a rack that rocks, slides, or tips under a correct wipe pass or a correct placement, or a shelf strip that will not hold a jar upright. Weight or replace it before the next episode.
- **Defective jar:** a jar that will not stand upright unaided, slips out of a correct grip, has a cap that will not stay tight, or carries a label that cannot be read on camera. Replace it before the next episode.
- **Defective cloth:** a cloth that is torn, shedding, or already loaded with dust at episode start.
- **Two heights too close to separate:** the jar set does not sort into three clearly different height groups by eye. Rebuild the set before the next episode.
- **A zone shifts out of the arc** during the episode, or a zone turns out to sit outside the arm's comfortable reach so a slot cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Move one jar from the rack to the staging area
2. Take the cloth from the cloth park
3. Wipe one shelf strip
4. Return the cloth to the cloth park
5. Move one jar within the staging area to sort it
6. Move one jar from the staging area to its slot on the rack
7. Face one jar forward
8. Return the arm home and end the episode